

CHAPTER 6 : SECOND LAW OF THERMODYNAMICS

Section	Core Revision Focus
6.1	Introduces the second law as the basis for determining process direction, feasibility, and the limitations of energy conversion.
6.2	Defines thermal energy reservoirs and distinguishes heat sources from heat sinks used in thermodynamic cycles.
6.3	Develops heat-engine operation, thermal efficiency, waste-heat rejection, and the Kelvin–Planck statement.
6.4	Covers refrigerators and heat pumps, COP, energy-efficiency measures, the Clausius statement, and their equivalence.
6.5	Explains perpetual-motion machines of the first and second kinds and the laws of thermodynamics they violate.
6.6	Distinguishes reversible and irreversible processes and identifies internal, external, total reversibility, and irreversibilities.
6.7	Develops the reversible Carnot cycle, its four processes, heat/work interactions, and maximum possible efficiency.
6.8	Establishes the Carnot principles: reversible engines are most efficient and all reversible engines between the same reservoirs have equal efficiency.
6.9	Develops the thermodynamic temperature scale and the relation between heat-transfer ratios and absolute temperatures.
6.10	Applies Carnot efficiency to heat engines and examines temperature limits, efficiency improvement, and energy quality/degradation.
6.11	Develops Carnot refrigerator and heat-pump COP relations and examines the effect of temperature limits on performance.

6.1 INTRODUCTION TO THE SECOND LAW

CORE IDEA

The first law gives **energy conservation**; the second law gives the **direction and limitations of processes**.

Feasible Process $\Rightarrow$ First Law Satisfied + Second Law Satisfied

LIMITATIONS OF THE FIRST LAW

- First law determines the **quantity** of energy and energy transformations.
- It does not determine the **direction** of a process.
- Natural processes have preferred directions; spontaneous reverse processes generally do not occur.
- The second law provides the criterion for process direction and theoretical performance limits.

ENERGY QUALITY

- First law $\rightarrow$ energy quantity/conservation.
- Second law $\rightarrow$ energy quality, work potential, degradation and process limitations.
- For equal quantities of energy, higher-temperature energy generally has greater work-producing potential.

Second Law $\rightarrow$ Direction + Energy Quality + Performance Limits

6.2 THERMAL ENERGY RESERVOIRS

THERMAL ENERGY RESERVOIR

A **thermal energy reservoir** is a body that can supply or absorb a finite amount of heat while maintaining essentially constant temperature.

$$\text{Thermal Energy Capacity} = mc$$

where m = mass and c = specific heat.

Reservoir condition: the thermal energy capacity is sufficiently large that the heat transfer causes negligible temperature change.

Typical models: oceans, lakes, rivers, atmospheric air, two-phase systems, controlled furnaces.

SOURCE AND SINK

Source $\longrightarrow$ Heat

A **source** supplies heat.

Heat $\longrightarrow$ Sink

A **sink** absorbs heat.

6.3 HEAT ENGINES

HEAT ENGINE

A **heat engine** is a cyclic device that receives heat from a high-temperature source, converts part of it into work, and rejects the remainder to a low-temperature sink.

$$Q_H = W_{\text{net,out}} + Q_L$$

where

$$Q_H = \text{heat supplied}, \quad Q_L = \text{heat rejected}$$

$$T_H > T_L$$

THERMAL EFFICIENCY

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_H}$$

Using the cycle energy balance,

$$W_{\text{net,out}} = Q_H - Q_L$$

hence

$$\eta_{\text{th}} = 1 - \frac{Q_L}{Q_H}$$

$$0 \leq \eta_{\text{th}} < 1$$

KELVIN--PLANCK STATEMENT

A cyclic device cannot receive heat from a single reservoir and produce a net amount of work. Therefore, a cyclic heat engine must interact with both a high-temperature and a low-temperature reservoir.

$$\eta_{\text{th}} < 1$$

Complete conversion of heat from a single reservoir into cyclic work is impossible.

WHY HEAT REJECTION IS NECESSARY

For continuous cyclic operation,

$$Q_H = W_{\text{net,out}} + Q_L$$

Thus some heat must be rejected to the low-temperature reservoir. Waste heat rejection is a **fundamental second-law requirement**, not merely a consequence of friction or imperfect construction.

6.4 REFRIGERATORS AND HEAT PUMPS

REFRIGERATOR

A refrigerator transfers heat from a low-temperature region to a high-temperature region using work input.

$$W_{\text{net,in}} = Q_H - Q_L$$

where Q_L is the refrigeration effect.

$$\text{COP}_R = \frac{Q_L}{W_{\text{net,in}}} = \frac{Q_L}{Q_H - Q_L} = \frac{1}{Q_H/Q_L - 1}$$

Note: COP_R may exceed unity; COP is not thermal efficiency.

HEAT PUMP

A heat pump uses work to transfer heat from a low-temperature source to a high-temperature space.

$$\text{COP}_{\text{HP}} = \frac{Q_H}{W_{\text{net,in}}} = \frac{Q_H}{Q_H - Q_L} = \frac{1}{1 - Q_L/Q_H}$$

REFRIGERATOR VS HEAT PUMP

Feature	Refrigerator	Heat Pump
Purpose	Cooling	Heating
Desired effect	Q_L	Q_H
Input	$W_{\text{net,in}}$	$W_{\text{net,in}}$
Performance	COP_R	COP_{HP}

Since

$$Q_H = Q_L + W_{\text{net,in}}$$

$$\text{COP}_{\text{HP}} = \text{COP}_R + 1$$

EER AND SEER

$$\text{EER} = \frac{\text{rate of heat removal}}{\text{rate of electricity consumption}} \quad [\text{Btu/Wh}]$$

$$\text{SEER} = \frac{\text{seasonal heat removed}}{\text{seasonal electricity consumed}} \quad [\text{Btu/Wh}]$$

$$\text{EER} = 3.412 \text{ COP}_R$$

CLAUSIUS STATEMENT

A cyclic device cannot produce no effect other than transferring heat from a lower-temperature body to a higher-temperature body.

A refrigerator can transfer heat from cold to hot only with external work input.

KELVIN--PLANCK VS CLAUSIUS

The two statements are **equivalent consequences** of the second law.

Violation of Kelvin-Planck $\Rightarrow$ Violation of Clausius

and conversely.

6.5 PERPETUAL-MOTION MACHINES

TYPES

Machine	Law Violated	Violation
PMM1	First law	Creates energy
PMM2	Second law	Violates process direction/energy-conversion limits

PMM1

Perpetual-motion machine of the first kind violates conservation of energy.

$$\text{PMM1} \Rightarrow \text{Energy Creation}$$

PMM2

A **PMM2** violates the second law without necessarily violating energy conservation.

A cyclic device that:

- receives heat from a single reservoir, and
- produces net work

is a PMM2.

ESSENTIAL DISTINCTION

First Law $\rightarrow$ Conservation

Second Law $\rightarrow$ Direction and Limitations

6.6 REVERSIBLE AND IRREVERSIBLE PROCESSES

REVERSIBLE PROCESS

A process is **reversible** if the system and surroundings can both be restored completely to their initial states with no net effect.

For the combined forward and reverse processes,

$$Q_{\text{net}} = 0, \quad W_{\text{net}} = 0$$

System restoration alone is insufficient; the surroundings must also be restored.

IRREVERSIBLE PROCESS

An **irreversible process** cannot be reversed without leaving a net change in the system or surroundings.

All natural processes are irreversible.

Reversible processes are ideal limiting models.

ENGINEERING IMPORTANCE

For work-producing devices:

$\text{Reversible} \rightarrow W_{\text{out,max}}$

For work-consuming devices:

$\text{Reversible} \rightarrow W_{\text{in,min}}$

Reversible operation therefore establishes the theoretical performance limit.

MAJOR IRREVERSIBILITIES

- friction,
- unrestrained expansion,
- mixing,
- heat transfer through a finite temperature difference,
- electrical resistance,
- inelastic deformation,
- chemical reactions.

TYPES OF REVERSIBILITY

Type	Condition
Internally reversible	No irreversibilities inside system boundary
Externally reversible	No irreversibilities outside system boundary
Totally reversible	No internal or external irreversibilities

Internally reversible: system passes through equilibrium states and forward/reverse paths coincide.

Externally reversible: heat transfer occurs without a finite temperature difference at the boundary.

$\text{Totally reversible} = \text{Internally reversible} + \text{Externally reversible}$

6.7 THE CARNOT CYCLE

CARNOT CYCLE

The Carnot cycle is a completely reversible cycle consisting of:

- two reversible isothermal processes,
- two reversible adiabatic processes.

It operates between

$$T_H > T_L$$

FOUR PROCESSES

Process	Type	Heat	Temperature
1 → 2	Isothermal expansion	Q_H in	$T_H = \text{constant}$
2 → 3	Adiabatic expansion	$Q = 0$	$T_H \rightarrow T_L$
3 → 4	Isothermal compression	Q_L out	$T_L = \text{constant}$
4 → 1	Adiabatic compression	$Q = 0$	$T_L \rightarrow T_H$

CYCLE RELATIONS

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$$

$$\boxed{W_{\text{net,out}} = Q_H - Q_L}$$

On the P–V diagram, the enclosed area represents net work output:

$$W_{\text{net,out}} = W_{\text{expansion}} - W_{\text{compression}}$$

MAXIMUM PERFORMANCE

Because the Carnot cycle is completely reversible,

$$\boxed{\eta_{\text{Carnot}} = \eta_{\text{rev}} = \eta_{\text{max}}}$$

for a heat engine operating between the same two temperature limits.

No actual engine operating between the same limits can exceed Carnot efficiency.

REVERSED CARNOT CYCLE

Reversing every process gives a Carnot refrigerator/heat pump:

$$Q_L : \text{absorbed}, \quad Q_H : \text{rejected}, \quad W_{\text{net,in}} : \text{supplied}$$

6.8 THE CARNOT PRINCIPLES

FIRST CARNOT PRINCIPLE

An irreversible heat engine is less efficient than a reversible heat engine operating between the same two reservoirs.

$$\eta_{th,irrev} < \eta_{th,rev}$$

Irreversibilities $\longrightarrow$ Lower efficiency

SECOND CARNOT PRINCIPLE

All reversible heat engines operating between the same two reservoirs have the same efficiency.

$$\eta_{rev,1} = \eta_{rev,2} = \eta_{rev,3}$$

Their efficiency depends only on the two reservoir temperatures, not on:

- working fluid,
- engine design,
- method of executing the cycle.

COMBINED RESULT

For engines operating between the same two reservoirs,

$$\eta_{irrev} < \eta_{rev,1} = \eta_{rev,2} = \eta_{rev,3}$$

The reversible engine establishes the **upper performance limit**.

6.9 THE THERMODYNAMIC TEMPERATURE SCALE

BASIC RESULT

For reversible heat engines, efficiency depends only on reservoir temperatures:

$$\eta_{th,rev} = g(T_H, T_L)$$

Therefore,

$$\left(\frac{Q_H}{Q_L}\right)_{rev} = f(T_H, T_L)$$

THREE-ENGINE RELATION

For three reversible engines,

$$\frac{Q_1}{Q_2} = f(T_1, T_2), \quad \frac{Q_2}{Q_3} = f(T_2, T_3), \quad \frac{Q_1}{Q_3} = f(T_1, T_3)$$

Since

$$\frac{Q_1}{Q_3} = \frac{Q_1}{Q_2} \frac{Q_2}{Q_3}$$

the functional form can be written as

$$f(T_1, T_2) = \frac{\phi(T_1)}{\phi(T_2)}$$

Hence,

$$\frac{Q_H}{Q_L} = \frac{\phi(T_H)}{\phi(T_L)}$$

for a reversible heat engine.

KELVIN TEMPERATURE SCALE

Choose

$$\phi(T) = T$$

Then

$$\left(\frac{Q_H}{Q_L}\right)_{rev} = \frac{T_H}{T_L}$$

or

$$\frac{Q_H}{Q_L} = \frac{T_H}{T_L} \quad (\text{reversible})$$

This defines the thermodynamic/Kelvin absolute-temperature scale.

IMPORTANT KELVIN-SCALE FACTS

- Absolute temperature scale.
- Independent of properties of a particular thermometric substance.
- Temperature ratios correspond directly to reversible heat-transfer ratios.
- Lower limit is absolute zero.

TRIPLE POINT AND CELSIUS SCALE

A fixed temperature reference is required to establish the temperature-unit magnitude. The source notes use the **triple point of water** as the fixed reference.

$$1\text{ K} \equiv 1^\circ\text{C}$$

$$T(^{\circ}\text{C}) = T(\text{K}) - 273.15$$

Temperature differences have equal numerical magnitudes in K and $^{\circ}\text{C}$.

ABSOLUTE ZERO

$$T \geq 0$$

Absolute zero is the lower limit of thermodynamic temperature.

PRACTICAL DETERMINATION

Absolute temperature need not be measured with an actual reversible heat engine. A **constant-volume ideal-gas thermometer** can be used for practical temperature measurement.

6.10 THE CARNOT HEAT ENGINE

CARNOT EFFICIENCY

For a reversible heat engine,

$$\eta_{th,Carnot} = 1 - \frac{T_L}{T_H}$$

Condition: T_H and T_L must be **absolute temperatures**.

$$\eta_{th,actual} < \eta_{th,Carnot}$$

and, in general,

$$\eta_{th} \leq \eta_{th,rev}$$

for a physically possible engine operating between the same temperature limits.

TEMPERATURE EFFECTS

From

$$\eta_{th,rev} = 1 - \frac{T_L}{T_H}$$

$$T_H \uparrow \Rightarrow \eta_{th,rev} \uparrow$$

$$T_L \downarrow \Rightarrow \eta_{th,rev} \uparrow$$

Thus, theoretically:

$$\boxed{\text{Increase } T_H \text{ and/or decrease } T_L}$$

to increase the maximum efficiency.

Practical limits arise from material temperature limits and available cooling temperatures.

DEPENDENCE OF CARNOT EFFICIENCY

Carnot efficiency depends only on

$$T_H, \quad T_L$$

and not on:

- working fluid,
- engine design,
- details of the reversible cycle,

- method of executing reversible processes.

QUALITY OF ENERGY

The first law concerns **quantity**; the second law also concerns **quality/work potential**.

Higher temperature $\Rightarrow$ Higher work-producing potential

All work can be completely converted to heat, but only part of heat can be converted to work.

Work $\rightarrow$ Heat (complete conversion possible)

Heat $\rightarrow$ Work (second-law limited)

ENERGY DEGRADATION

When heat flows from high temperature to low temperature:

Energy quantity remains conserved

but

Work-producing potential decreases

Thus,

Energy degradation = reduction in work-producing potential
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6.11 THE CARNOT REFRIGERATOR AND HEAT PUMP

CARNOT REFRIGERATOR

For a refrigerator operating reversibly between T_H and T_L ,

$$W_{\text{net,in}} = Q_H - Q_L$$

$$\text{COP}_{R,\text{rev}} = \frac{Q_L}{Q_H - Q_L}$$

Using

$$\frac{Q_H}{Q_L} = \frac{T_H}{T_L}$$

gives

$$\text{COP}_{R,\text{rev}} = \frac{1}{T_H/T_L - 1} = \frac{T_L}{T_H - T_L}$$

CARNOT HEAT PUMP

For a reversible heat pump,

$$\text{COP}_{\text{HP},\text{rev}} = \frac{Q_H}{Q_H - Q_L}$$

and therefore

$$\text{COP}_{\text{HP},\text{rev}} = \frac{1}{1 - T_L/T_H} = \frac{T_H}{T_H - T_L}$$

TEMPERATURE EFFECTS ON REFRIGERATOR COP

$$\text{COP}_{R,\text{rev}} = \frac{T_L}{T_H - T_L}$$

At fixed T_H ,

$$T_L \downarrow \Rightarrow \text{COP}_{R,\text{rev}} \downarrow$$

and as

$$T_L \rightarrow 0, \quad \text{COP}_R \rightarrow 0$$

Hence, refrigeration becomes increasingly difficult as the refrigerated space approaches absolute zero.

TEMPERATURE EFFECTS ON HEAT-PUMP COP

$$\text{COP}_{\text{HP,rev}} = \frac{T_H}{T_H - T_L}$$

A larger temperature lift,

$$T_H - T_L \uparrow$$

requires more work and therefore

$$\text{COP}_{\text{HP,rev}} \downarrow$$

Hence, heat-pump performance improves when the required temperature difference is reduced.

RELATIONSHIP BETWEEN COPs

$$\text{COP}_{\text{HP}} = \text{COP}_{\text{R}} + 1$$

and for reversible operation,

$$\text{COP}_{\text{HP,rev}} = \text{COP}_{\text{R,rev}} + 1$$

Therefore,

$$\text{COP}_{\text{HP}} > \text{COP}_{\text{R}}$$

for the same operating conditions.

REVERSIBLE VS IRREVERSIBLE DEVICES

For the same temperature limits:

Device	Second-Law Limit
Irreversible refrigerator	$\text{COP}_{\text{R}} < \text{COP}_{\text{R,rev}}$
Reversible refrigerator	$\text{COP}_{\text{R}} = \text{COP}_{\text{R,rev}}$
Impossible refrigerator	$\text{COP}_{\text{R}} > \text{COP}_{\text{R,rev}}$
Irreversible heat pump	$\text{COP}_{\text{HP}} < \text{COP}_{\text{HP,rev}}$
Reversible heat pump	$\text{COP}_{\text{HP}} = \text{COP}_{\text{HP,rev}}$
Impossible heat pump	$\text{COP}_{\text{HP}} > \text{COP}_{\text{HP,rev}}$

The reversible values are the **maximum possible COPs** between the specified temperature limits.